

Panvee Naidu

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SUMMARY

Final-year Computer Science undergraduate (Cybersecurity specialization) who builds and ships full-stack systems end to end. Experience spans real-time services with Socket.IO, Redis and PostgreSQL, Docker-orchestrated multi-container deployments, JWT-authenticated REST APIs in FastAPI and Node, MQTT publish/subscribe messaging, and PKCS#11 hardware-security-module cryptography. Comfortable from message brokers and container orchestration through to accessible, performant front ends. Seeking a remote software engineering internship.

EDUCATION

SRM University – AP

B.Tech, Computer Science and Engineering — Specialization in Cybersecurity

Expected May 2027

CGPA: 7.4/10

EXPERIENCE

Salesforce Developer Intern

May 2025 – Jul 2025

SmartBridge — Agentblazer Champion Program, in collaboration with Salesforce

Remote

- Completed an 8-week structured development program covering organizational setup, data modeling, security management, and process automation.
- Built and tested custom **Apex** classes and **Lightning Web Components**, applying object-oriented and event-driven patterns; earned the **Object Relationships Superbadge** and certified as an **Agentblazer Champion**.

PROJECTS

AeroChat — Real-Time Chat Platform

github.com/panveenaidu/aero-chat-realttime · React 18, Node.js, Express, Socket.IO, PostgreSQL, Redis, Docker Compose, JWT

- Built a **real-time messaging platform** with Socket.IO delivering instant messages across multiple rooms, with typing indicators, join/leave events, and full message history persisted to **PostgreSQL**.
- Architected for **horizontal scaling** using a **Redis** adapter to synchronise Socket.IO events across multiple backend instances, with message and user state surviving container restarts via persistent volumes.
- Implemented **JWT authentication with bcrypt** password hashing (10 rounds), 24-hour token expiry, and authenticated WebSocket connections — securing both the REST API and the real-time channel.
- Containerised the full stack with **Docker Compose** — frontend, backend, PostgreSQL and Redis as orchestrated services — with a Makefile, deployment script, and a health-check endpoint for readiness probes.

Aurum — Hotel Management Platform

github.com/panveenaidu/aurum-hotel-management · Python 3.13, FastAPI, SQLAlchemy, React 18, TypeScript, JWT, Tailwind CSS

- Engineered a **role-based multi-tenant platform** with three isolated portals — platform admin, hotel admin and guest — enforced by **JWT role guards** at the router layer so each role can only reach its own endpoints.
- Designed a layered **FastAPI** backend separating routers, Pydantic schemas, SQLAlchemy tables and a services layer for business logic, with automatic OpenAPI documentation and bcrypt-hashed credentials.
- Built a **commission engine** that calculates and records platform earnings automatically on booking confirmation, plus hotel approval workflows, room inventory, and guest booking lifecycle management.

PKCS#11 OTP Authentication System

github.com/panveenaidu/pkcs11-otp-authentication · C, PKCS#11, SoftHSM, Apache, PHP, MySQL

- Implemented **hardware-security-module-backed OTP authentication** using the **PKCS#11** cryptographic token standard with **SoftHSM**, generating and verifying one-time passwords with keys held inside the token rather than in application memory.
- Integrated the cryptographic layer into a booking application on an Apache/PHP/MySQL stack, with token initialisation, slot configuration, and OS-level group permissions controlling HSM access.

Smart Home Automation — Distributed IoT System

github.com/panvee/Home-Automation-Using-Voice-Command · Python, MQTT (paho-mqtt), REST APIs, Multi-threading, Mosquitto

- Architected a **distributed IoT platform across three decoupled processes** over **MQTT publish/subscribe**, with a multi-threaded REST API and a voice interpreter routing **13 intent patterns** to handlers.
- Identified and documented **security gaps in own design during post-build review** — unauthenticated REST endpoints behind an authenticated UI, and unencrypted MQTT transport — with a prioritized remediation plan.

Valorix — VALORANT Performance Platform

github.com/panveenaidu/valorix · Next.js 15, React 19, TypeScript, Tailwind CSS v4, Framer Motion, Vercel

- Architected a production **Next.js 15 / React 19** application with a custom Tailwind v4 design-token system, hand-built SVG charts with **zero charting dependencies**, **prefers-reduced-motion** accessibility guards on every animation, and a GitHub → Vercel CI/CD pipeline under strict TypeScript.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, Apex, SQL, HTML, CSS

Backend & Systems: FastAPI, Node.js, Express, REST API design, Socket.IO / WebSockets, MQTT publish-subscribe, Redis, PostgreSQL, MySQL, SQLAlchemy, multi-threaded and event-driven services

Frontend: React 18/19, Next.js, Vite, TypeScript, Tailwind CSS, Framer Motion, Zustand, TanStack Query, accessible UI

DevOps & Security: Docker, Docker Compose, CI/CD, Linux, Git, GitHub, Vercel, JWT and bcrypt authentication, PKCS#11, SoftHSM, Mosquitto

AI & Practices: YOLOv8, OpenCV, NumPy; distributed system design, security review, code review

CERTIFICATIONS & ACHIEVEMENTS

SAP Certified — SAP Generative AI Developer, SAP (2026) • **GenAI for Front-End Developers**, Coursera (2025) • **Intro to Front-End Development**, Meta (2025) • **Google AI Essentials**, Google (2025)

Participant, **Data Analytics** and **Data Science with Python** Hackathons — TechGyan, with Elan & Nvision, **IIT Hyderabad** (2025)